Implementation Plan

Project 05 — Dual-Core RV32I SoC with Simplified Coherent Memory Subsystem

Document ID: IMPL-RISCV-SOC-05
Owner: arafi
Timeline: 5 days (Sep 1–5, 2026)
PDK: sky130A | **Flow:** OpenLane | **Bus:** AXI4-Lite | **FPGA:** Arty A7-100T / Zybo

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1. Purpose

This is the **execution document**. It breaks the 5-day timeline into granular tasks with:

- **What to do** (the task)
- **Dependencies** (what must be done first)
- **Deliverable** (the concrete output)
- **Acceptance check** (how to know it's correct)
- **Time budget** (how many hours)

If the PRD defines *what* and the TRD defines *how*, this document defines *when and in what order*.

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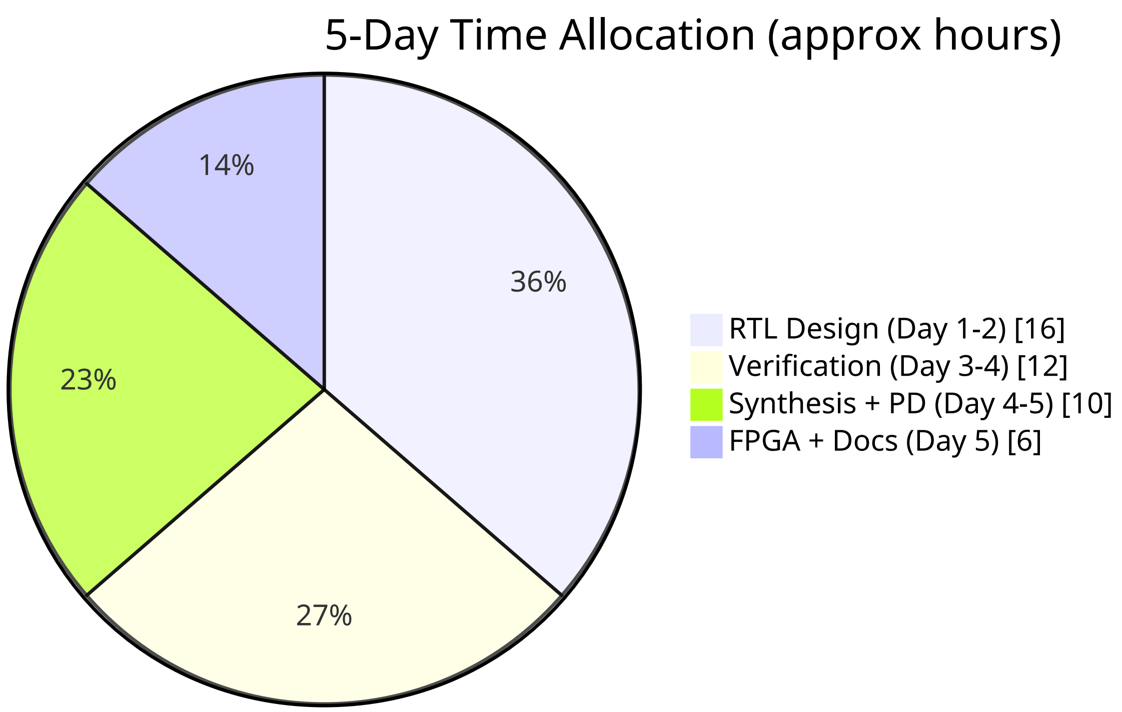
2. Time Budget Overview



Day	Focus	Hours	Key Risk
Day 1 (Sep 1)	Logic design + core RTL	8	CPU bugs
Day 2 (Sep 2)	Cache + coherence + bus RTL	8	Coherence FSM bugs
Day 3 (Sep 3)	Top integration + directed TB + synth smoke	8	Integration bugs, synth failures
Day 4 (Sep 4)	UVM + OpenLane physical design	8	UVM scope creep, PD issues
Day 5 (Sep 5)	FPGA + sign-off + documentation	6	FPGA pin/timing, doc time

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3. Day 1 — Logic Design + Core RTL

3.1 Morning (4 hours): Logic Design on Paper

Task	ID	Dependencies	Deliverable	Acceptance	Hours
Design ALU logic table	T1.1	—	ALU operation truth table	All RV32I ops defined with select encoding	0.5
Design D-cache FSM state diagram	T1.2	—	FSM diagram on paper	All states, transitions, reset values defined	1.0
Design coherence controller FSM	T1.3	T1.2	FSM diagram on paper	I/S/M transitions + invalidation path defined	0.5
Design round-robin arbiter	T1.4	—	Logic table	2-master arbiter truth table	0.5
Draw system block diagram + memory map	T1.5	T1.1-T1.4	Block diagram + memory map	All modules, connections, addresses shown	1.0
Finalize Day-1 decisions (see below)	T1.6	—	Decision record	7 decisions documented	0.5

Day-1 decisions to finalize (from analysis):

1. Core: single-cycle (confirmed)
2. Bus: AXI4-Lite (confirmed)
3. Write policy: write-through (confirmed)
4. Coherence: I/S/M 3-state (confirmed)
5. Cache: 4 lines, direct-mapped, 1 word/line
6. Reset: async active-low, single domain
7. SRAM: register arrays for I-SRAM; shared SRAM as register array (avoid OpenRAM integration cost)

3.2 Afternoon (4 hours): Core RTL

Task	ID	Dependencies	Deliverable	Acceptance	Hours
Write `alu.sv`	T1.7	T1.1	`rtl/core/alu.sv`	Compiles, simple test in TB	0.5
Write `reg_file.sv`	T1.8	—	`rtl/core/reg_file.sv`	2R/1W, x0=0, compiles	0.5
Write `control_unit.sv`	T1.9	T1.1	`rtl/core/control_unit.sv`	Decodes all RV32I instructions	1.0
Write `rv32i_core.sv`	T1.10	T1.7-T1.9	`rtl/core/rv32i_core.sv`	Integrates ALU+RF+CU+PC+data IF	1.0
	T1.11	T1.10			1.0

Task	ID	Dependencies	Deliverable	Acceptance	Hours
Write simple core testbench			`tb/tb_core_smoke.sv`	Executes a few instructions (ADD, ADDI, SW, LW), checks reg values	

Day 1 Exit Criteria

- [ ] ALU, reg_file, control_unit, rv32i_core all compile
- [ ] Core executes at least: ADDI, ADD, LW, SW, BEQ correctly in sim
- [ ] All 7 design decisions documented
- [ ] FSM diagrams on paper (keep these — you'll reference them)

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4. Day 2 — Cache + Coherence + Bus RTL

4.1 Morning (4 hours): Memory + Cache

Task	ID	Dependencies	Deliverable	Acceptance	Hours
Write `sram_reg_array.sv` (generic)	T2.1	—	`rtl/memory/sram_reg_array.sv`	Parameterized depth/width, sync R/W	0.5
Instantiate `i_sram.sv` using generic	T2.2	T2.1	`rtl/memory/i_sram.sv`	1KB private, read-only port	0.25
Instantiate `shared_sram.sv` using generic	T2.3	T2.1	`rtl/memory/shared_sram.sv`	4KB shared, 1 RW port	0.25
Write `d_cache.sv` (line storage)	T2.4	T1.2	`rtl/cache/d_cache.sv`	4 lines, valid/tag/data/state fields	0.5
Write `d_cache_mgr.sv` (FSM)	T2.5	T2.4, T1.2	`rtl/cache/d_cache_mgr.sv`	FSM implements hit/miss/write/invalidate, AXI-Lite master port	2.0
Test d_cache_mgr in isolation	T2.6	T2.5	`tb/tb_cache_smoke.sv`	Hit returns data, miss fetches via AXI, write updates SRAM	0.5

4.2 Afternoon (4 hours): Coherence + Bus + Peripherals

Task	ID	Dependencies	Deliverable	Acceptance	Hours
Write `coherence_ctrl.sv`	T2.7	T1.3	`rtl/coherence/coherence_ctrl.sv`	Tracks I/S/M, dispatches invalidation on write_notify	1.0
Test coherence_ctrl in isolation	T2.8	T2.7	`tb/tb_coh_smoke.sv`	Write from core 0 invalidates core 1 line	0.5
Write `axi_lite_arbiter.sv`	T2.9	T1.4	`rtl/bus/axi_lite_arbiter.sv`	Round-robin, 2 masters, deadlock-free	0.5
Write `axi_lite_decoder.sv`	T2.10	—	`rtl/bus/axi_lite_decoder.sv`	Routes to SRAM/MMIO/UART/GPIO, DECERR on unmapped	0.5

Task	ID	Dependencies	Deliverable	Acceptance	Hours
Write `mmio_regs.sv`	T2.11	—	`rtl/peripheral/mmio_regs.sv`	AXI-Lite slave, status/counters/doorbell/control	0.5
Write `uart_core.sv`	T2.12	—	`rtl/peripheral/uart_core.sv`	AXI-Lite slave, 115200 8N1 TX/RX	0.75
Write `gpio_led.sv`	T2.13	—	`rtl/peripheral/gpio_led.sv`	AXI-Lite slave, LED output reg	0.25

Day 2 Exit Criteria

- [ ] All memory, cache, coherence, bus, peripheral modules compile
- [ ] d_cache_mgr passes isolation test (hit/miss/write)
- [ ] coherence_ctrl passes isolation test (invalidation works)
- [ ] AXI arbiter + decoder compile and route correctly

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5. Day 3 — Top Integration + Directed TB + Synth Smoke

5.1 Morning (4 hours): Integration + Directed TB

Task	ID	Dependencies	Deliverable	Acceptance	Hours
Write `riscv_soc_top.sv`	T3.1	all RTL	`rtl/top/riscv_soc_top.sv`	Instantiates and wires everything; compiles	1.5
Write AXI-Lite interface definitions	T3.2	—	`rtl/bus/axi_lite_if.sv`	SV interface for all AXI-Lite connections	0.5
Write `tb_directed.sv` test cases 1-5	T3.3	T3.1	`tb/directed/tb_directed.sv`	Reset, single-core access, hit, miss, cross-core coherence	1.5
Debug integration	T3.4	T3.3	working sim	First 5 directed tests pass	0.5

5.2 Afternoon (4 hours): More Tests + Synth Smoke

Task	ID	Dependencies	Deliverable	Acceptance	Hours
Write directed test cases 6-10	T3.5	T3.4	extended `tb_directed.sv`	Stale prevention, simultaneous write, no-cached-copy, error, counters	1.0
Debug all directed tests	T3.6	T3.5	all 10 tests pass	Self-checking, PASS flag at end	1.0
Write demo programs (core0/core1 .hex)	T3.7	T3.4	`sw/core0.hex`, `sw/core1.hex`	Simple asm that does the coherence demo scenario	0.5
Yosys synthesis smoke test	T3.8	T3.1	Yosys log	Synthesizes with 0 errors, 0 latches, all mapped	1.0
Fix any synth issues	T3.9	T3.8	clean synth	Re-synth until clean	0.5

Day 3 Exit Criteria

- [ ] `riscv_soc_top` integrated and compiles

- [ ] All 10 directed test cases pass (self-checking)
- [ ] Demo program hex files generated
- [ ] Yosys synthesis smoke test passes (0 errors, 0 latches)
- [ ] This is the critical gate — if synth is clean here, Day 5 is much easier

> **Transparency note:** Day 3 is the pivot day. If the directed tests don't pass, do NOT proceed to UVM or physical design. Fix the core behavior first. A passing directed TB is your truth source.

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6. Day 4 — UVM + OpenLane Physical Design

6.1 Morning (4 hours): UVM Environment

Task	ID	Dependencies	Deliverable	Acceptance	Hours
Write `axi_lite_if.sv` for UVM	T4.1	T3.2	`tb/uvm/axi_lite_if.sv`	SV interface for UVM connection	0.25
Write `axi_lite_seq_item.sv`	T4.2	—	transaction object	addr, data, we, wmask, response	0.25
Write `axi_lite_master_agent.sv`	T4.3	T4.2	driver + sequencer + monitor	Drives AXI-Lite transactions	1.0
Write sequences (core0, core1, coherence_vseq)	T4.4	T4.3	sequences	Randomized read/write to shared addrs	0.75
Write `shared_sram_scoreboard.sv`	T4.5	T4.3	scoreboard	Predicts SRAM state, compares with monitor	0.75
Write `coherence_monitor.sv`	T4.6	—	monitor	Observes invalidation events	0.5
Write `coverage_collector.sv`	T4.7	—	covergroups	Coherence states, hit/miss, inv count, error	0.5

6.2 Afternoon (4 hours): UVM Run + OpenLane

Task	ID	Dependencies	Deliverable	Acceptance	Hours
Assemble UVM env + test	T4.8	T4.1-T4.7	`tb/uvm/tb_uvm.sv`	Env instantiates all components	0.5
Run UVM, debug	T4.9	T4.8	UVM logs	1000+ random txns pass, scoreboard matches	1.5
Generate coverage report	T4.10	T4.9	coverage report	Functional >=90%, code >=90% line	0.5
Write SDC constraints	T4.11	—	`constraints/soc.sdc`	Clock 50ns, I/O delays defined	0.25
Write OpenLane `config.json`	T4.12	T3.8	`openlane/config.json`	Design name, files, clock, PDK, utilization	0.25
Run OpenLane full flow	T4.13	T4.11, T4.12	OpenLane output	Synth + floorplan + PDN + place + CTS + route + sign-off	1.0

Day 4 Exit Criteria

- [ ] UVM environment runs, scoreboard passes 1000+ random txns
- [ ] Coverage report generated (functional + code)
- [ ] SDC written
- [ ] OpenLane config written
- [ ] OpenLane flow runs to completion (may have DRC/timing issues to fix Day 5)

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7. Day 5 — Sign-off + FPGA + Documentation

7.1 Morning (4 hours): Sign-off + FPGA

Task	ID	Dependencies	Deliverable	Acceptance	Hours
Check DRC (Magic)	T5.1	T4.13	DRC report	0 violations (fix if needed)	0.5
Check LVS (Netgen)	T5.2	T4.13	LVS report	Clean netlist match	0.5
Check timing (STA)	T5.3	T4.13	timing report	WNS >= 0 (setup + hold)	0.25
Capture layout screenshot (KLayout)	T5.4	T5.1	layout PNG	GDS visual screenshot	0.25
Write FPGA `.xdc` constraints	T5.5	—	`constraints/fpga.xdc`	Pins, clock, I/O standards	0.5
Add MMCM wrapper for FPGA	T5.6	—	`rtl/top/fpga_top.sv`	MMCM generates 50 MHz from board ref	0.5
Vivado synth + impl + bitstream	T5.7	T5.5, T5.6	`.bit` file	No negative slack, bitstream generated	1.0
Program board + run demo	T5.8	T5.7	live demo / video	UART shows coherence, LEDs blink	0.5

7.2 Afternoon (2-3 hours): Documentation + Cleanup

Task	ID	Dependencies	Deliverable	Acceptance	Hours
Write `architecture.md`	T5.9	all	`docs/architecture.md`	Covers all modules, coherence, memory map, results	1.0
Organize reports (synth, PD, coverage)	T5.10	T4.10, T5.1-T5.4	organized report files	All reports in `reports/` folder	0.5
Final deliverables checklist	T5.11	all	checklist	All PRD deliverables checked	0.5

Day 5 Exit Criteria

- [ ] DRC: 0 violations
- [ ] LVS: clean
- [ ] Timing: WNS >= 0
- [ ] Layout screenshot captured
- [ ] FPGA bitstream generated
- [ ] Board demo runs (UART + LEDs show coherence)
- [ ] Architecture document complete

- [ ] All deliverables checked off

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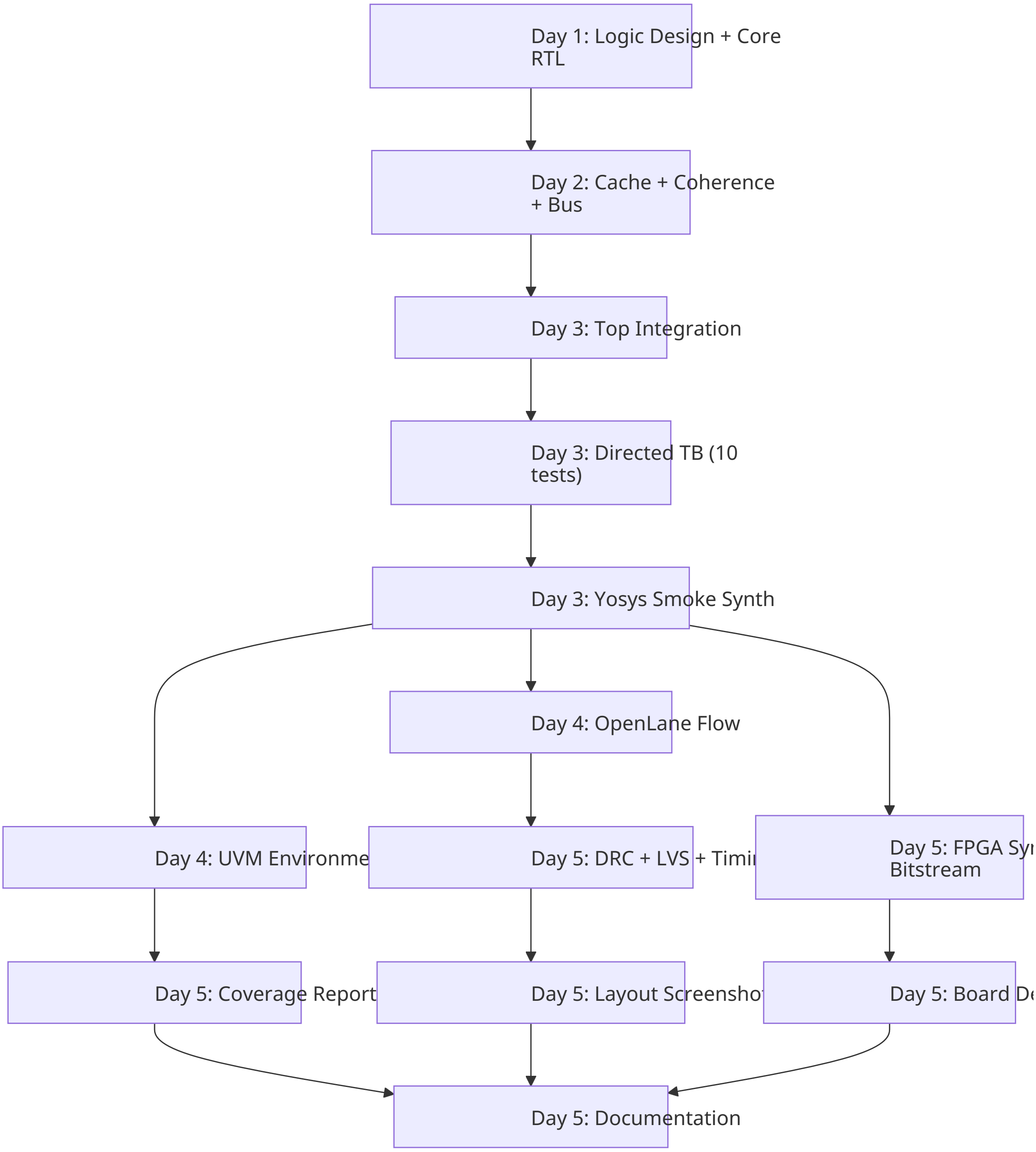
8. Task Dependency Graph (Critical Path)



Critical path: T1 → T2 → T3a → T3b → T3c → T4b → T5b → T5f.
Any slip on this path slips the whole project.

Parallelizable on Day 4: UVM (T4a) and OpenLane (T4b) can run in parallel if you have two terminals/ machines.

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9. Daily Standup Checklist (use these each morning)

Day 1 Standup

- [ ] Do I have a known-good RV32I core design to reference?
- [ ] Have I finalized all 7 design decisions?
- [ ] Is my toolchain ready (simulator, Yosys, Vivado, RISC-V GCC)?

Day 2 Standup

- [ ] Is the core from Day 1 passing its smoke test?
- [ ] Do I have the FSM diagrams from Day 1 to reference?
- [ ] Am I keeping the coherence mechanism simple (3-state, write-through)?

Day 3 Standup

- [ ] Are all modules individually tested before integration?
- [ ] Am I running the Yosys smoke test today (not skipping it)?

Day 4 Standup

- [ ] Is the directed TB fully passing before I start UVM?
- [ ] Is the OpenLane config ready to run?
- [ ] Am I keeping UVM minimal (not over-engineering)?

Day 5 Standup

- [ ] Are DRC/LVS/timing clean or do I have a plan to fix?
- [ ] Are the FPGA constraints correct for my board?
- [ ] Do I have time allocated for documentation?

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10. Risk Contingency Plan

If this happens...	Then...
Core doesn't work by end of Day 1	Use a simpler known-good open-source single-cycle core; don't debug from scratch.
Directed TB fails on Day 3	Stop. Fix the RTL. Do not proceed to UVM/PD until directed tests pass.
Yosys synth fails on Day 3	Fix unsynthesizable constructs (latches, initial blocks, $display). This is a hard gate.
OpenLane DRC violations on Day 4-5	Usually floorplan/routing. Adjust utilization, re-run. If persistent, simplify the design (fewer cache lines).
Timing doesn't close (WNS < 0)	Lower clock to 25 MHz (40 ns). If still failing, check critical path — likely ALU or cache tag compare. Consider multicycle data path as last resort.
UVM doesn't work by end of Day 4	Prioritize: keep the directed TB (it's the required baseline). Submit UVM partially working with coverage report from what ran.
FPGA doesn't work on Day 5	If Vivado timing fails, lower core clock to 25 MHz. If pin issues, double-check board schematic. If all else fails, show behavioral sim demo as fallback.
Run out of time for docs	Architecture doc is required. Write a minimal version (block diagram + module list + coherence explanation + results). Better minimal than missing.

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11. Deliverables Tracking Matrix

Deliverable	Task ID	Day	Status
`alu.sv`	T1.7	1	[ ]
`reg_file.sv`	T1.8	1	[ ]
`control_unit.sv`	T1.9	1	[ ]
`rv32i_core.sv`	T1.10	1	[ ]
`sram_reg_array.sv`	T2.1	2	[ ]
`i_sram.sv`	T2.2	2	[ ]
`shared_sram.sv`	T2.3	2	[ ]
`d_cache.sv`	T2.4	2	[ ]
`d_cache_mgr.sv`	T2.5	2	[ ]
`coherence_ctrl.sv`	T2.7	2	[ ]
`axi_lite_arbiter.sv`	T2.9	2	[ ]
`axi_lite_decoder.sv`	T2.10	2	[ ]
`mmio_regs.sv`	T2.11	2	[ ]
`uart_core.sv`	T2.12	2	[ ]
`gpio_led.sv`	T2.13	2	[ ]
`riscv_soc_top.sv`	T3.1	3	[ ]
`axi_lite_if.sv`	T3.2	3	[ ]
Directed TB (10 tests)	T3.3-T3.6	3	[ ]
Demo programs (.hex)	T3.7	3	[ ]
Yosys synth clean	T3.8-T3.9	3	[ ]
UVM env	T4.1-T4.8	4	[ ]
UVM run + coverage	T4.9-T4.10	4	[ ]
SDC constraints	T4.11	4	[ ]
OpenLane config	T4.12	4	[ ]
OpenLane full flow	T4.13	4	[ ]
DRC clean	T5.1	5	[ ]
LVS clean	T5.2	5	[ ]
Timing met	T5.3	5	[ ]
Layout screenshot	T5.4	5	[ ]
FPGA .xdc	T5.5	5	[ ]
FPGA MMCM wrapper	T5.6	5	[ ]
FPGA bitstream	T5.7	5	[ ]
Board demo	T5.8	5	[ ]
Architecture doc	T5.9	5	[ ]
Reports organized	T5.10	5	[ ]

Deliverable	Task ID	Day	Status
Final checklist	T5.11	5	[ ]

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12. Definition of Done (Project Complete When ALL Are True)

- [ ] All RTL modules written, compile, and are synthesizable (0 Yosys errors, 0 latches)
- [ ] Directed self-checking TB: all 10 test cases pass
- [ ] UVM: scoreboard passes 1000+ random transactions
- [ ] Coverage: functional >=90%, code >=90% line
- [ ] OpenLane flow completes: DRC 0 violations, LVS clean
- [ ] Timing: WNS >= 0 (setup and hold met)
- [ ] Reports captured: area, timing, power, congestion, DRC, LVS
- [ ] Layout screenshot captured (KLayout)
- [ ] FPGA: bitstream generated, board running
- [ ] FPGA demo: UART shows coherence sequence, LEDs show activity
- [ ] Architecture document delivered
- [ ] All deliverables from PRD checked off

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13. Quick Reference - Key Files

File	Purpose	Created Day
`rtl/core/rv32i_core.sv`	RV32I single-cycle CPU	1
`rtl/cache/d_cache_mgr.sv`	Cache controller FSM + AXI master	2
`rtl/coherence/coherence_ctrl.sv`	Coherence controller	2
`rtl/top/riscv_soc_top.sv`	Top-level integration	3
`tb/directed/tb_directed.sv`	Self-checking directed TB	3
`tb/uvm/tb_uvm.sv`	UVM testbench top	4
`constraints/soc.sdc`	ASIC timing constraints	4
`openlane/config.json`	OpenLane configuration	4
`constraints/fpga.xdc`	FPGA pin constraints	5
`rtl/top/fpga_top.sv`	FPGA top with MMCM	5
`docs/architecture.md`	Architecture document	5